If the camp leader distributes 6 litres of water in $\frac{1}{2}$ litre water bottles to the students, then how many students will get water bottles? This means finding how many $\frac{1}{2}$ litres are there in 6 litres. For this we need to calculate $6 \div \frac{1}{2}$.

Solution Let us describe the situation

Amount of water	Picture	Amount of water distributed	Method of finding it	Number of persons received
6 litres		1 litre	$6 \div 1$	6
6 litres		$\frac{1}{2}$ litre	$6 \div \frac{1}{2}$	12
6 litres		$\frac{1}{4}$ litre	$6 \div \frac{1}{4}$	24

This means that if you share 6 litres of water into 1 litre bottles, 6 persons can get water. If you share in $\frac{1}{2}$ litre water bottles, 12 persons can get water. If you share it in $\frac{1}{4}$ litre water bottles, 24 persons can get water. That is

$6 \div 1 = 6$	$6 \div \frac{1}{2} = 12$	$6 \div \frac{1}{4} = 24$

We can illustrate this in the following diagram. We divide each circle into halves such that each part is $\frac{1}{2}$ of the whole. The number of such halves would be $6 \div \frac{1}{2}$. In the figure how many halves do you see? There are 12 halves. So $6 \div \frac{1}{2} = 6 \times 2 = 12$.



As one circle has 2 halves, 6 circles will have 12 halves = 6×2. Therefore, $6 \div \frac{1}{2} = 6 \times 2 = 12$

6th_ Maths _Chapter 1.indd 16 22-10-2019 18:12:39